

# Second-Order Pooling for Protein Language Models

## 1 Motivation

Protein language models (pLMs) like ESM-2, ProtT5, and ProtX (distilled ProtT5) produce rich per-residue embeddings, but most downstream tasks require a single fixed-size vector per protein. The standard choice in the field is *mean pooling*: average all residue embeddings into one vector, then feed that to a probe. It is simple, stable, and surprisingly competitive which is why essentially every pLM benchmark pipeline uses it as the default.

However, mean pooling has a structural blind spot: it averages each feature dimension independently, which destroys information about feature *co-occurrence*. If two features are biologically meaningful only when they appear together at the same residue for example, a specific motif occurring inside an accessible loop, or two cysteines that form a disulfide bond mean pooling collapses these into the same representation as a protein where those features occur at different positions.

A recent paper from Goodfire Research proposes an alternative for genomic foundation models: *covariance pooling*, which uses the second moment of residue activations rather than the first. This captures how features co-activate within a sequence, preserving combinatorial structure that mean pooling discards. They report substantial gains on gene-level tasks when applied to the NTV3 genomic model.

The idea of this project is to test whether this approach transfers to protein language models. To our knowledge, covariance/second-order pooling has not been applied to pLMs despite a 10-year history of strong results in computer vision (bilinear CNNs, compact bilinear pooling) and the recent positive results in genomics.

## 2 Research Question

Does covariance-based pooling of per-residue ProtX embeddings improve per-protein downstream task performance compared to mean pooling, and at what embedding size does it become parameter-efficient?

## Hypothesis

Covariance pooling should preserve residue feature co-occurrence structure that mean pooling destroys, yielding improved downstream performance potentially at comparable or smaller embedding sizes than the original pLM hidden dimension.

## 3 Method

### 1 Mathematical Formulation

Given per-residue embeddings  $\mathbf{X} \in \mathbb{R}^{L \times d}$  from a frozen pLM:

**Mean pooling (baseline):**

$$\boldsymbol{\mu} = \frac{1}{L} \mathbf{X}^\top \mathbf{1}_L \in \mathbb{R}^d \quad (1)$$

**Covariance pooling (this project):** learn projections  $\mathbf{L}, \mathbf{R} \in \mathbb{R}^{d \times d_c}$ , then compute

$$\mathbf{C} = \frac{1}{L} (\mathbf{X}\mathbf{L})^\top (\mathbf{X}\mathbf{R}) \in \mathbb{R}^{d_c \times d_c} \quad (2)$$

The resulting  $d_c \times d_c$  matrix is flattened to a  $d_c^2$ -dimensional vector and fed to the same probe head as mean pooling.

Two training regimes for the projections:

- **Supervised:** train  $\mathbf{L}, \mathbf{R}$  end-to-end with the probe head.
- **Unsupervised:** train  $\mathbf{L}, \mathbf{R}$  by reconstructing the full covariance matrix  $\mathbf{X}^\top \mathbf{X}$ , exploiting the Frobenius-norm equivalence  $\|\mathbf{X}^\top \mathbf{X}\|_F = \|\mathbf{X}\mathbf{X}^\top\|_F$  so no sequence-by-sequence matrix is materialized. Freeze and reuse across tasks.

**Important:** proper masking of padded positions is critical for both mean and covariance pooling. Implement and unit-test them side-by-side.

## 4 Experimental Setup

### 1 Backbone

- **Model:** ProtX (provided), frozen.
- **Layer:** last-layer per-residue embeddings.
- **Caching:** extract embeddings once using the provided inference script; save to disk (HDF5). All subsequent experiments run on cached tensors.

### 2 Framework

Biotrainer using the `residues_to_class` and `residues_to_value` protocols these protocols take per-residue embeddings and require pooling inside the model, which is exactly where our intervention lives.

### 3 Tasks

Classification + regression gives breadth. If time allows, binary membrane/soluble classification can be added as a cheap third task.

| Task                      | Protocol          | Type                    | Dataset         | Metric         |
|---------------------------|-------------------|-------------------------|-----------------|----------------|
| Subcellular Localization  | residues_to_class | 10-class classification | DeepLoc / FLIP  | Accuracy / Q10 |
| Meltome (thermostability) | residues_to_value | Regression              | FLIP<br>Meltome | Spearman R     |

## 4 Pooling Methods

| Method                                    | Parameters            | Embedding size |
|-------------------------------------------|-----------------------|----------------|
| Mean pooling (baseline)                   | 0                     | $d$            |
| Covariance pooling, supervised            | $2 \cdot d \cdot d_c$ | $d_c^2$        |
| Covariance pooling, unsupervised (frozen) | $2 \cdot d \cdot d_c$ | $d_c^2$        |
| Hybrid: $[\mu; \text{flat}(\mathbf{C})]$  | $2 \cdot d \cdot d_c$ | $d + d_c^2$    |

All methods use the same Biotrainer probe head to ensure a fair, apples-to-apples comparison.

## 5 Experiments

### 1 Core Experiments

- **Head-to-head benchmark:** 4 pooling methods  $\times$  2 tasks  $\times$  3 seeds = 24 main results.
- **Size-matched efficiency sweep:** vary  $d_c \in \{8, 16, 24, 32, 48\}$ , plot downstream metric vs. embedding size.
- **Supervised vs. unsupervised bottleneck:** train the unsupervised autoencoder once, freeze, reuse across tasks; compare against per-task supervised bottlenecks.
- **Layer sweep:** repeat the head-to-head on a handful of layers (last, last 4, middle, early) does covariance pooling's advantage hold across layers?

### 2 Stretch Goals (if core finishes early)

- Light Attention and attention pooling baselines.
- Pool PaRTI (PageRank-based pooling).
- Binary membrane/soluble classification as a third task.

## 6 Plots and Visualizations

A deliberate visualization plan to make results clear and compelling.

## 1 Minimum Viable (required)

- **Main results bar chart:** per task, 4 pooling methods side-by-side with error bars from 3 seeds.
- **Size vs. performance efficiency curve:** X: embedding dim (log scale); Y: downstream metric. Covariance pooling as a curve, mean pooling as a horizontal line. Vertical dashed line at the equal-size crossover.
- **Layer sweep heatmap:** rows: pooling method; columns: layer; cells: downstream metric. One heatmap per task, shared color scale.

## 2 Target Delivery (what the project aims for)

- **Covariance matrix visualization:** pick 2–3 representative proteins (e.g., one per subcellular class), show their  $d_c \times d_c$  learned covariance embeddings as heatmaps. Pair with the corresponding mean-pooled vectors shown as 1D heatmaps to visually contrast what information mean pooling discards.
- **Feature co-activation analysis:** for one task, identify the top-weighted entries  $(i, j)$  in the learned covariance embedding, and highlight which residues in a sample protein have the largest  $(\mathbf{XL})_{k,i} \cdot (\mathbf{XR})_{k,j}$  products. If possible, compare top residues to annotated functional/structural sites.
- **t-SNE / UMAP clustering:** project flattened covariance embeddings of all test proteins to 2D, color by class label (for SCL). Show mean-pooled embeddings side-by-side for comparison.

## 7 References

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